

OpenNutri AI / Algorithm Master Defense

Midterm exposition deck | Retrieval, ranking, bilingual search, and feedback reuse

Role framing

Retrieval design across multiple scientific sources

Ranking logic before PDF acquisition

EN/TR language split and Turkish-source support

Feedback reuse from annotation outcomes

Main midterm claim

OpenNutri already operates as a closed loop linking search, filtering, PDF acquisition, annotation, and feedback reuse.

This deck explains that loop from the AI / algorithm side, then goes deeper into the implementation details likely to matter in the defense.

Why this layer exists

The algorithmic problem is not merely collecting papers; it is deciding which papers deserve download and expert time.

01

Distributed literature

Relevant food-composition papers are spread across Europe PMC, OpenAlex, Semantic Scholar, DergiPark, and local open-access paths. A single-source strategy misses too much.

02

High acquisition cost

Blindly downloading every candidate PDF is expensive, noisy, and wasteful. The system needs ranking before acquisition, not only after download.

03

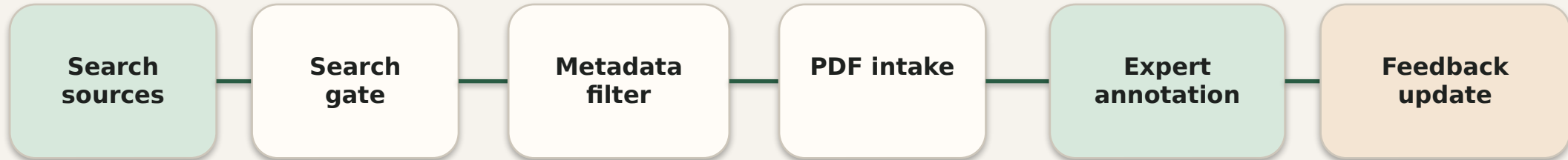
Turkish visibility gap

Turkish studies are underrepresented in standard international food-data systems. Retrieval therefore needs explicit EN/TR separation and Turkish-source support.

Goal of the AI layer: maximize useful paper flow to the annotator while preserving recall and avoiding unnecessary PDF work.

What OpenNutri is at midterm

For the AI role, the important point is that the project is one connected loop rather than a separate scraper, backend, and UI.



What it is

A human-in-the-loop scientific paper discovery system

A shared data contract between crawler, storage, and annotation

A feedback-driven retrieval system that changes later runs

What it is not

Not a one-shot scraper that downloads everything

Not only a database schema or only a labeling UI

Not yet a fully automatic extraction engine

End-to-end architecture

The architecture already links discovery, storage, annotation, and later feedback reuse. This is the system map for the rest of the presentation.

How to read this figure

Search sources produce metadata candidates.

The crawler decides which candidates deserve acquisition.

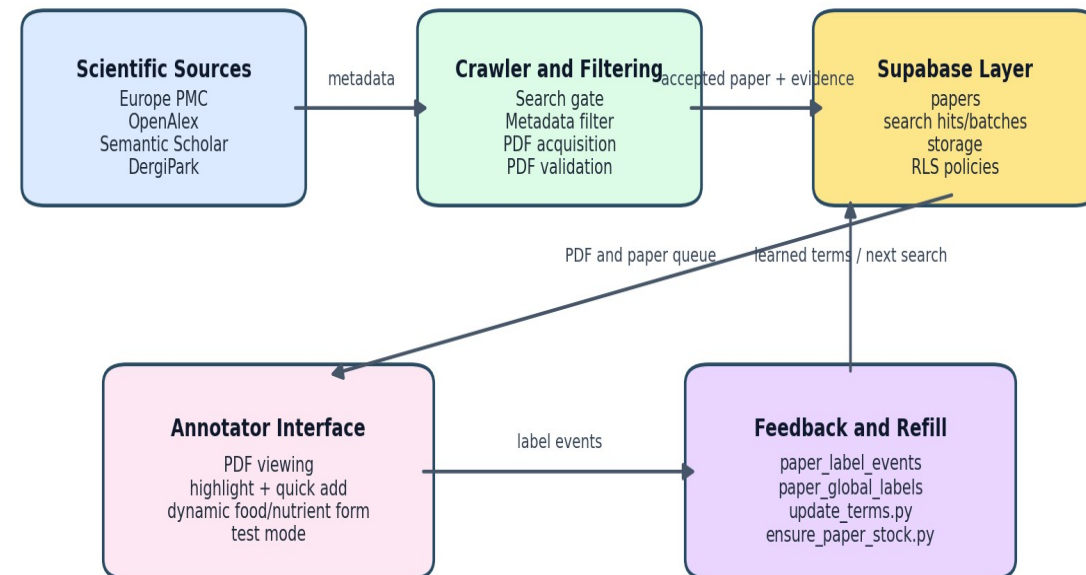
Accepted papers move into Supabase storage and paper records.

The annotator creates structured food-item and nutrient evidence.

Label events and global labels later feed the next retrieval cycle.

Main defense point: the AI layer is connected to real user decisions, not isolated from the product.

OpenNutri Midterm Architecture



Closed-loop logic: expert labels are not only a final output; they also influence later search and ranking decisions.

Shared data model relevant to AI

The retrieval layer depends on papers, search evidence, user events, and a shared vocabulary.

Reference vocabulary: entities, aliases, and master nutrients are reused by both retrieval and annotation.

Paper intake: papers store accepted PDF-level records available to the UI.

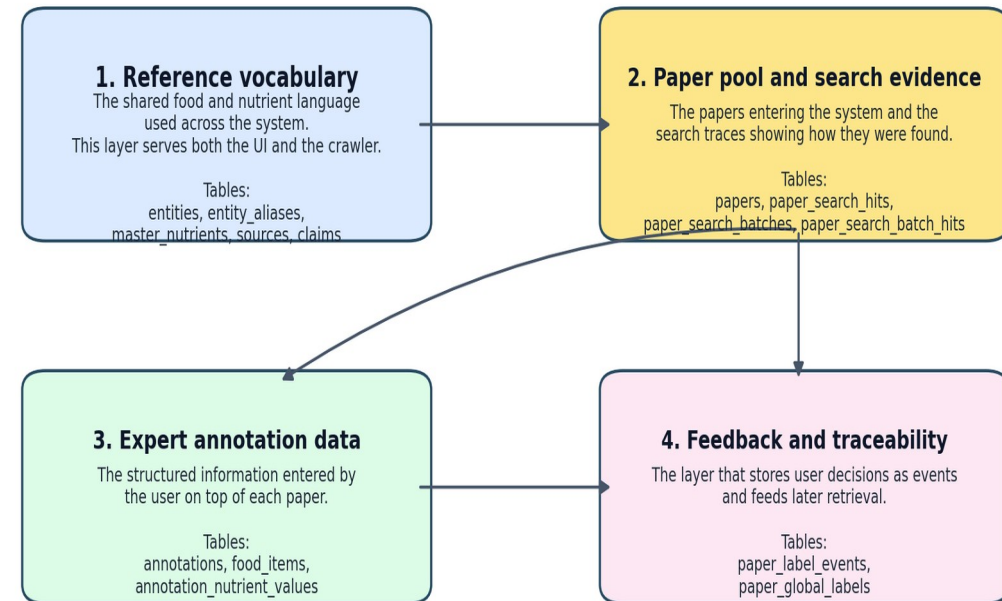
Search evidence: `paper_search_hits`, `paper_search_batches`, and `paper_search_batch_hits` preserve why a paper entered the system.

Feedback evidence: `paper_label_events`, `paper_global_labels`, and `search_sessions` preserve user behavior for later ranking updates.

Important distinction: raw search evidence is stored separately from accepted paper records.

OpenNutri Database Structure (Simplified)

The UI and crawler operate on the same Supabase schema



This is not a full ER diagram. It groups the core tables defined in `migration.sql` into four responsibilities so the project is easier to understand at a glance.

Search layer and language split

The crawler is multi-source by design, and it treats English and Turkish as different retrieval environments rather than one merged pool.

Europe PMC

Main open-access biomedical surface; strongest default bias for English.

OpenAlex

Broad metadata coverage used in both languages.

Semantic Scholar

Additional discovery path and duplicate-aware evidence.

DergiPark

Essential Turkish source; highest static bias in Turkish.

EN

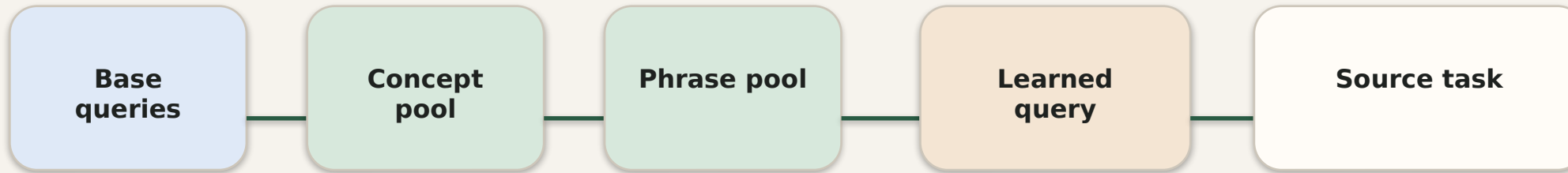
Default bias favors Europe PMC, then OpenAlex.
English composition frames and phrase pools stay language-scoped.

TR

Default bias favors DergiPark, then OpenAlex and Semantic Scholar.
Separate anchors, phrases, and feedback scores keep Turkish behavior language-scoped.

Query construction

The crawler does not rely on a fixed query list. It mixes base composition queries, learned query phrases, and a rotating concept pool.



Start with language-specific base composition templates.
Interleave food terms and nutrient terms in one concept pool.
Reorder concepts using `concept_scores` learned from previous outcomes.
Mix core phrases with a small exploration slice from the longer phrase pool.

Exact implementation choices

Exploration rate for extra phrases: 2 percent of the phrase pool.
Separate cursors are preserved per language.
Nutrient learned queries add unit framing such as mg/100g or g/100g.
Queries are normalized and deduplicated before search-task expansion.

Search-task prioritization

After query construction, each query is expanded into source-specific search tasks and ranked before execution.

Priority score

$$\text{priority_score} = \text{pair_score} + \text{batch_score}$$

$\text{pair_score} = \text{learned pair feedback} + 0.15 \times \text{source_prior} + \text{static source bias}$

batch_score comes from historical performance of the exact batch key

Tasks are sorted descending and truncated to the run budget

Static source bias by language

EN:

Europe PMC +0.35

OpenAlex +0.15

Semantic Scholar +0.05

DergiPark -0.15

TR:

DergiPark +0.35

OpenAlex +0.20

Semantic Scholar +0.10

Europe PMC -0.30

This is how the system turns language strategy into execution order.

Search gate

The search gate is a cheap first-pass metadata screen. It rejects weak candidates before the heavier metadata filter and acquisition stages.

Positive signals

- First anchor/composition phrase hit (+0.90)
- First food term hit (+0.35)
- First nutrient term hit (+0.35)
- Unit pattern such as mg/100g or g/100g (+0.70)
- Food + nutrient combo bonus (+0.45)

Penalties and pass rule

- Missing metadata -> immediate reject with score -2.0
- Missing abstract penalty: -1.30 or -0.45
- Strong negative signals penalized up to -2.5
- Soft negatives penalized up to -1.5
- Health-outcome terms penalized up to -1.6
- Pass threshold: score ≥ -0.35

Metadata filter

The metadata filter is the stronger additive decision layer. It combines lexical cues, source priors, embeddings, and feedback terms before PDF acquisition.

Core evidence

Composition phrase +1.35

Unit pattern +1.25

Food hit +0.65 and nutrient hit +0.65

Food + nutrient combo +0.75

Missing abstract penalty -1.10 or -0.35

Health-outcome penalty capped at -2.0

Learned and semantic evidence

Source prior from historical positive/negative balance, clamped to +/-0.9

Embedding similarity adds +1.45 if similarity crosses the language threshold

An extra +0.75 is added when embedding similarity is above threshold

Feedback n-gram score is summed and clamped to +/-6.0

Accept threshold: score ≥ 1.75

Embedding layer

Embeddings are a semantic support layer inside metadata scoring, not a standalone classifier.

English model

all-MiniLM-L6-v2
threshold = 0.45

Turkish / multilingual model

paraphrase-multilingual-MiniLM-L12-v2
threshold = 0.42

Shared rule

Trim metadata text to 1800 characters before encoding.

How the score is produced

Load anchor phrases from feedback config; fall back to seeded anchors if needed.

Encode the candidate title + abstract in the language-specific model.

Compute normalized similarity against all anchors and keep the maximum match.

Why this design

Anchors keep the semantic target narrow: composition-oriented rather than general nutrition.

Two models keep EN and TR scoring language-aware.

This layer recovers semantically relevant phrasing that exact keywords may miss.

Feedback labels and training sets

The learning loop is based on the latest visible state of each annotator, not on raw event totals.

Positive paper: latest draft or done state with `has_data = true`, food-item count > 0, and nutrient-value count > 0.

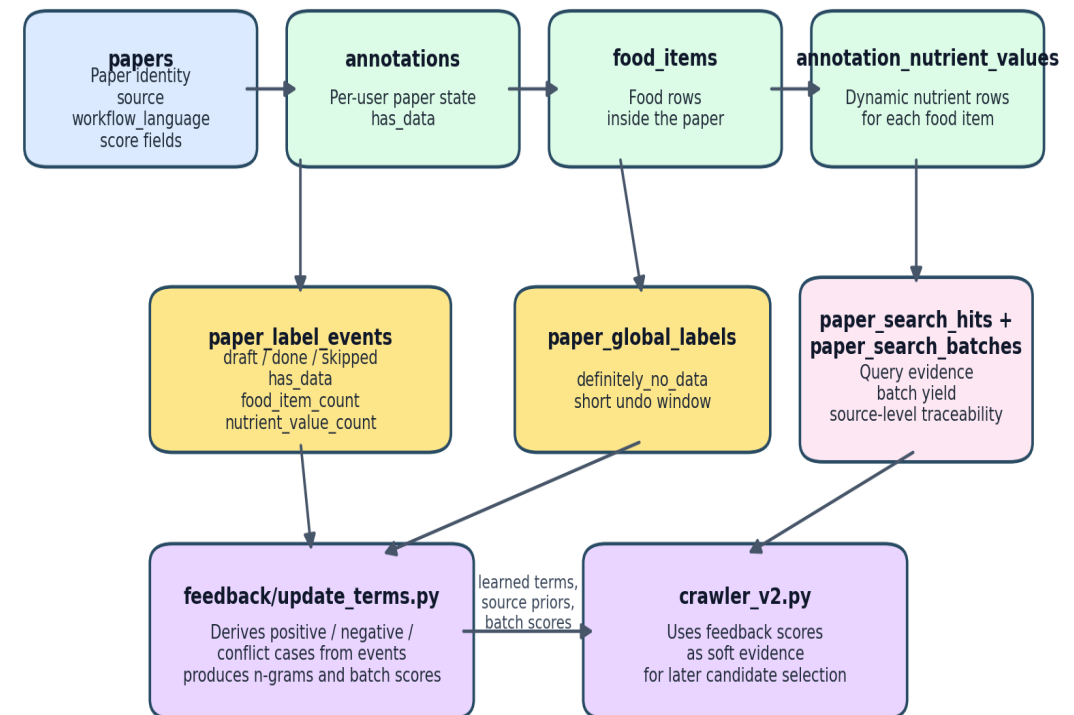
Negative paper: a global definitely-no-data label or at least two unique skip users.

Conflict paper: both positive and negative evidence; excluded from learning.

Processing is language-scoped after label construction.

Important: the feedback loop is implemented, but it is batch-updated rather than online.

Annotation and Feedback Data Relations



What feedback changes in later runs

The loop is already active. Feedback is not only stored; it is exported into a new config and consumed by the next crawler run.



query_phrases
anchor_phrases
weighted_terms
concept_scores
pair_scores, batch_scores, source_priors

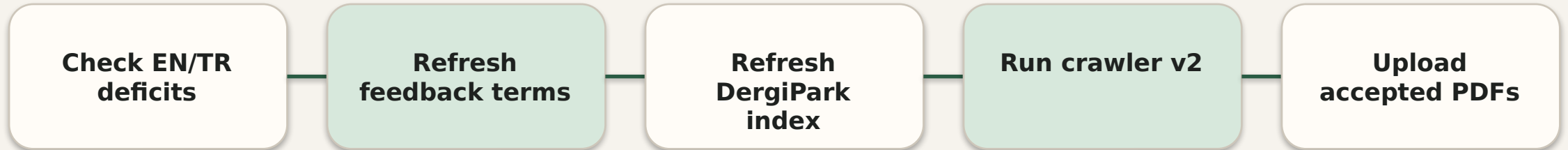
Later runs search different phrases in different orders.

Concept ordering changes which foods or nutrients are combined first.

Source-query combinations with better yield move up the queue.

Operational closure

The refill script is what turns the algorithmic pieces into a working operational loop when the annotator queue runs low.



The system can react when paper stock falls below target.
Feedback refresh is part of the refill cycle by default.
DergiPark refresh keeps the Turkish pool renewable.

The strongest sentence here is: “The current system is operational, not only experimental: it can refresh feedback, search new papers, and push accepted PDFs back into the annotator pipeline.”

AI-relevant UI interaction

The frontend is not only data entry. It captures structured evidence that the retrieval layer can later learn from.

PDF highlight and click resolution

Nutrient names are matched with boundary-aware regex patterns, wrapped in marks, and opened through direct-target, elementsFromPoint, or caret-based fallback when PDF.js text layers behave irregularly.

Quick-add popover and structured save

A clicked nutrient opens a popover near the highlight, captures value and unit, and adds a structured nutrient row into the current food-item form.

Autocomplete ranking and telemetry

Food and nutrient searches use custom ranking rules, and search_sessions record typed queries, shown options, selected options, and abandon cases.

Current state and scope boundary

This slide is for staying technically strong without overclaiming.

Implemented now

- Multi-source search across Europe PMC, OpenAlex, Semantic Scholar, and DergiPark
- Search gate plus stronger metadata filter before acquisition
- Dual-model embedding support inside metadata scoring
- Batch-updated feedback loop that changes later runs
- Operational refill loop tied to paper stock

Not claimed at midterm

- No trained classifier for query ranking yet
- No LLM-based search-term ranking
- No fully automatic extraction pipeline in the current loop
- Document segmentation and extraction are deferred to the second semester

Defense takeaway

If you had to compress the whole AI story into a few claims, these are the ones worth defending strongly.

Claim 1

The system performs staged filtering before acquisition. This is the main reason expert time is not wasted on every retrieved paper.

Claim 2

English and Turkish are separate retrieval pools. This is a real technical decision, not only a presentation detail, and DergiPark is part of that strategy.

Claim 3

The feedback loop is implemented now. It updates phrases, weighted terms, and source/query priorities for later runs.

If challenged, come back to these three claims and then answer with one concrete implementation detail.

Appendix: implementation detail and Q&A

The next slides are designed for long committee questions.

Appendix: implementation detail and Q&A

The appendix goes deeper into schema slices, exact thresholds, feedback rules, UI internals, and safe answers for common committee questions.

Appendix A. Sources and language strategy

This is the detailed justification for using multiple sources and for not merging EN and TR into one ranking pool.

Europe PMC provides open-access biomedical coverage and is the strongest English default surface.

OpenAlex broadens metadata coverage and supports both languages when PMC coverage is weak.

Semantic Scholar adds another discovery path and duplicate-aware evidence.

DergiPark is the essential Turkish source and justifies the language-specific Turkish workflow.

Different source visibility

Different term surfaces and composition phrasing

Different source priors and static biases

Different anchor and query-phrase pools

Appendix B. Exact weights and thresholds

These are the implemented constants for the two main ranking stages.

Search gate

composition phrase	+0.90
food term	+0.35
nutrient term	+0.35
unit pattern	+0.70
food + nutrient combo	+0.45
missing metadata	-2.00 reject
missing abstract	-1.30 or -0.45
strong negatives	up to -2.50
soft negatives	up to -1.50
health terms	up to -1.60
pass if score $\geq$	-0.35

Metadata filter

composition phrase	+1.35
unit pattern	+1.25
food hit	+0.65
nutrient hit	+0.65
food + nutrient combo	+0.75
embedding above threshold	+1.45
embedding positive bonus	+0.75
source prior clamp	+/-0.90
feedback term clamp	+/-2.50
feedback score clamp	+/-6.00
accept if score $\geq$	1.75

Appendix C. Feedback scoring detail

This slide combines exact label rules with the term-ranking logic used by the feedback updater.

Positive = latest draft/done with has_data = true
and real structured rows

Negative = global definitely-no-data or at least two
unique skips

Conflict = both positive and negative evidence,
excluded from learning

```
query_rank = 1.75*title_good + 0.75*title_net +  
0.35*ta_net - max(0, title_bad)  
anchor_rank = 1.25*title_good + 1.0*ta_good +  
0.5*title_net + 0.25*ta_net - max(0, title_bad)
```

n-grams are extracted separately from titles and
title+abstract text

positive, negative, and background document
frequencies are compared with log-odds style scoring

query_phrases default max = 64, anchor_phrases
default max = 16

weighted_terms carries the full scored-term payload
into later metadata scoring

Appendix D. Pair, batch, and concept feedback

The feedback exporter learns more than phrases: it also learns which source-query combinations and concept terms produce better outcomes.

Pair feedback

Group search hits by source + template_id + source_term.
Score = novel_positive_count / retrieved.
Also derive source_priors as $\log((\text{positive} + 1) / (\text{negative} + 1))$.

Batch feedback

Group by batch_key over paper_search_batches and paper_search_batch_hits.
Score = positive_count / retrieved.

Concept feedback

Group by source_term.
Score = (positive_count - negative_count) / retrieved.
This later reorders the concept pool.

Appendix E. Acquisition and validation

Even after metadata acceptance, papers still need a robust PDF path and a full-text validation step before becoming accepted records.



Try OA XML and direct PDF links first.

Use package extraction or HTML-discovered links when direct links fail.

Use curl / fallback paths for unstable source behavior.

Extract full text with pdftotext.

Check for composition framing, table/method evidence, nutrient-unit signals, and food markers.

Reject candidates that still fail full-text usefulness checks.

Appendix F. UI internals relevant to AI

These details matter when the committee asks how human interaction is turned into structured signals.

PDF highlight

Build boundary-aware nutrient regex patterns, scan PDF.js text spans after render, wrap matches in mark tags, and resolve overlaps so shorter nutrient names do not break longer ones.

Click resolution

Try the direct event target first, then fall back to `elementsFromPoint`, then to `caretPositionFromPoint` or `caretRangeFromPoint` because PDF.js text layers do not behave like normal HTML.

Autocomplete and telemetry

Food and nutrient autocomplete use custom ranking rules, and `search_sessions` logs typed queries, shown options, selected options, status, and timestamps.

Appendix G. Audited sample run summary

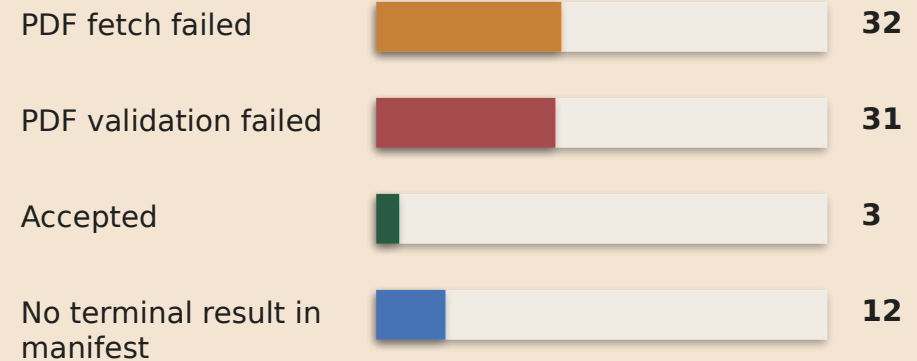
This slide intentionally separates stage counts from recorded acquisition outcomes so the sample run is not misread as a strict funnel.

Stage counts (TR live run 2026-03-30)



These counts come from the manifest summary and are valid as stage counts up to metadata_pass.

Recorded acquisition outcomes



Important audit note: 78 metadata-pass candidates exist, but only 66 have recorded terminal results in this manifest. Therefore this sample must not be presented as a strict funnel.

Appendix H. Likely questions and safe answers

These are the short answers to return to under pressure.

Q: Is the feedback loop implemented?

A: Yes. It is batch-updated, not online, and it already changes later crawler runs.

Q: Do you use an LLM to rank search terms?

A: No. Search-term ranking is feedback-driven statistical scoring, not LLM ranking.

Q: Did you train a model for query ranking?

A: Not yet. The current system learns through feedback statistics and embeddings, not a trained classifier.

Q: Why not download everything?

A: Because metadata filtering is much cheaper and protects expert time.

Q: Why separate English and Turkish?

A: Because source coverage, terminology, and source priors differ.

Q: Why not only embeddings?

A: Domain cues like units, composition phrases, and food/nutrient combinations are too valuable to ignore.